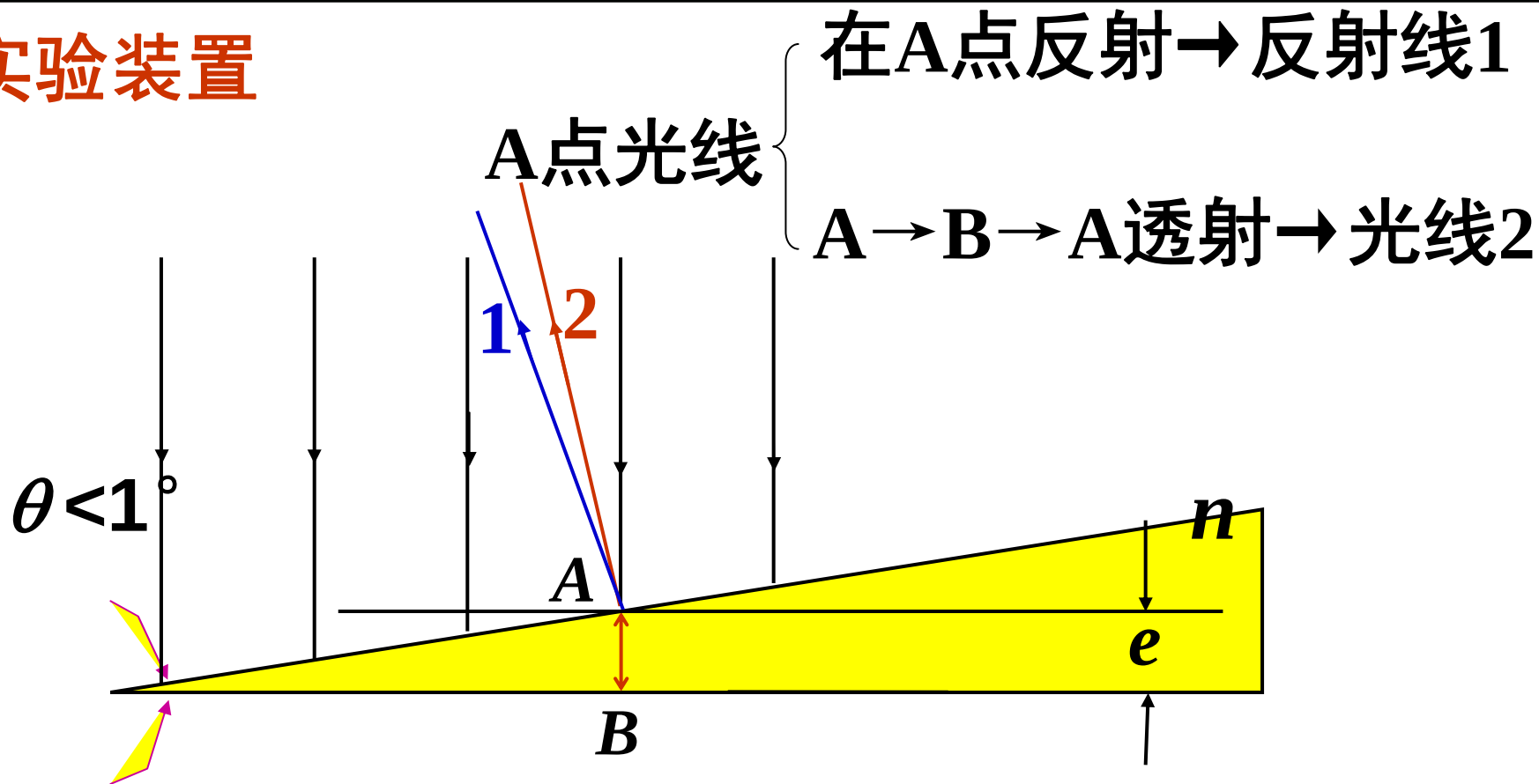


3.3.2 劈尖干涉

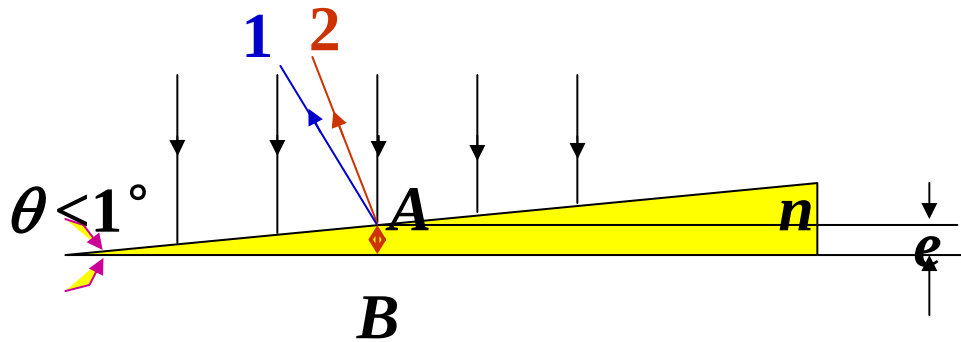
实验装置



干涉条纹分析

$$\delta = 2ne + \frac{\lambda}{2} \quad \delta(e) = \begin{cases} = k\lambda, & k = 1, 2, 3, \dots & \text{明纹} \\ = (2k + 1)\frac{\lambda}{2}, & k = 0, 1, 2, \dots & \text{暗纹} \end{cases}$$

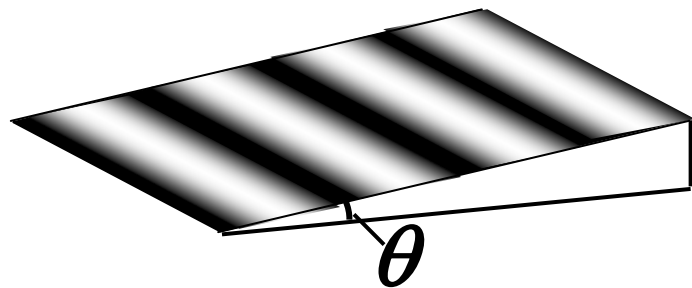
同一厚度 e 对应同一级条纹 —— 等厚条纹



讨论:

1) 平行光入射 $i = 0$,

$$\delta = 2ne + \frac{\lambda}{2}$$



与劈尖棱平行的直线上的各点 e 相同，在一个干涉级上，干涉花样为与劈尖棱平行的等间距的直线条纹

$$\delta = 2ne + \frac{\lambda}{2}$$

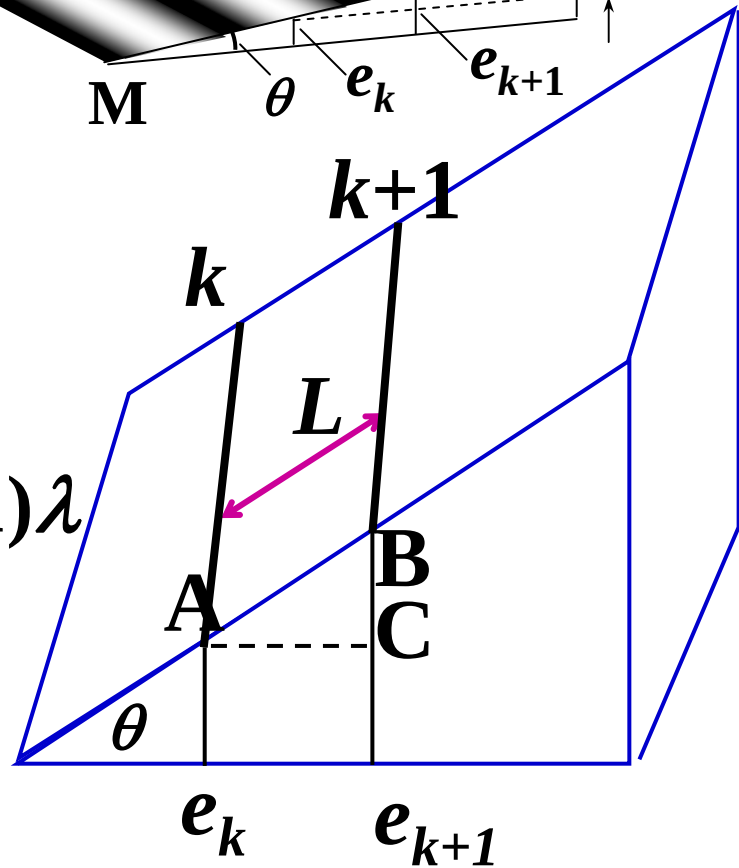
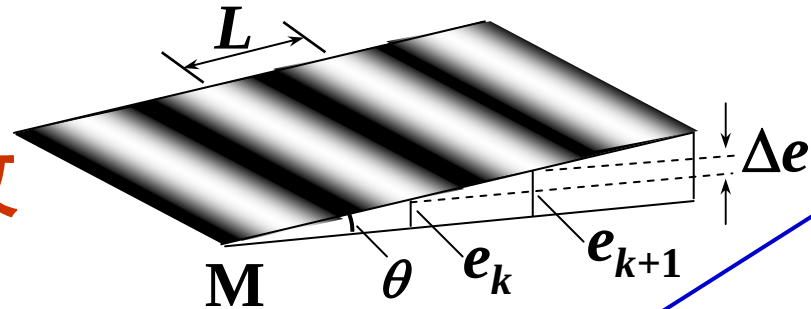
2) 棱边 $e=0$ $\delta = \lambda/2$ 暗纹

3) 相邻明（暗）纹间距 L

$$k \text{ 级明纹 } 2ne_k + \frac{\lambda}{2} = k\lambda$$

$$(k+1) \text{ 级明纹 } 2ne_{k+1} + \frac{\lambda}{2} = (k+1)\lambda$$

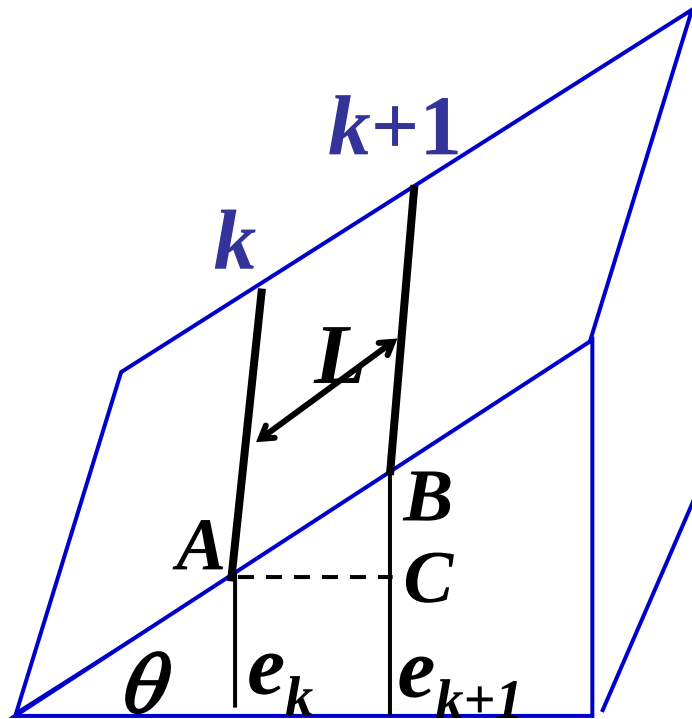
$$e_{k+1} - e_k = \frac{\lambda}{2n}$$



$$e_{k+1} - e_k = \frac{\lambda}{2n}$$

$$L = \frac{\lambda}{2n \sin \theta}$$

$$L = \frac{\lambda}{2n \theta}$$



$$L = \frac{\lambda}{2n\theta}$$

(1) L 与 k 无关，条纹等间距

(2) $L \propto \lambda$

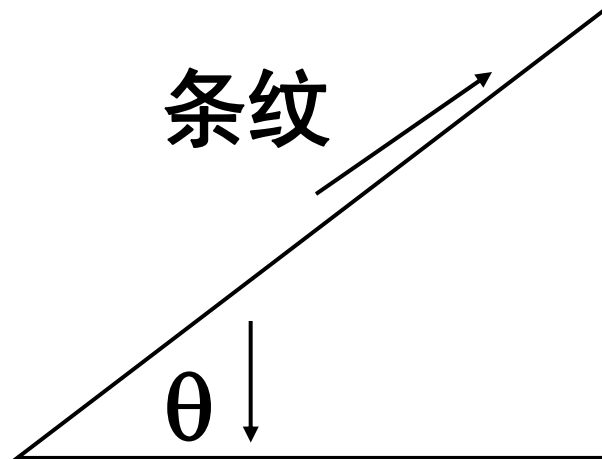
(3) $L \propto \frac{1}{\theta}$ $\theta \uparrow, L \downarrow$ 条纹变密

(4) $L \propto \frac{1}{n}$ n 越大条纹变密

4) 条纹动态分析

$$L = \frac{\lambda}{2n\theta}$$

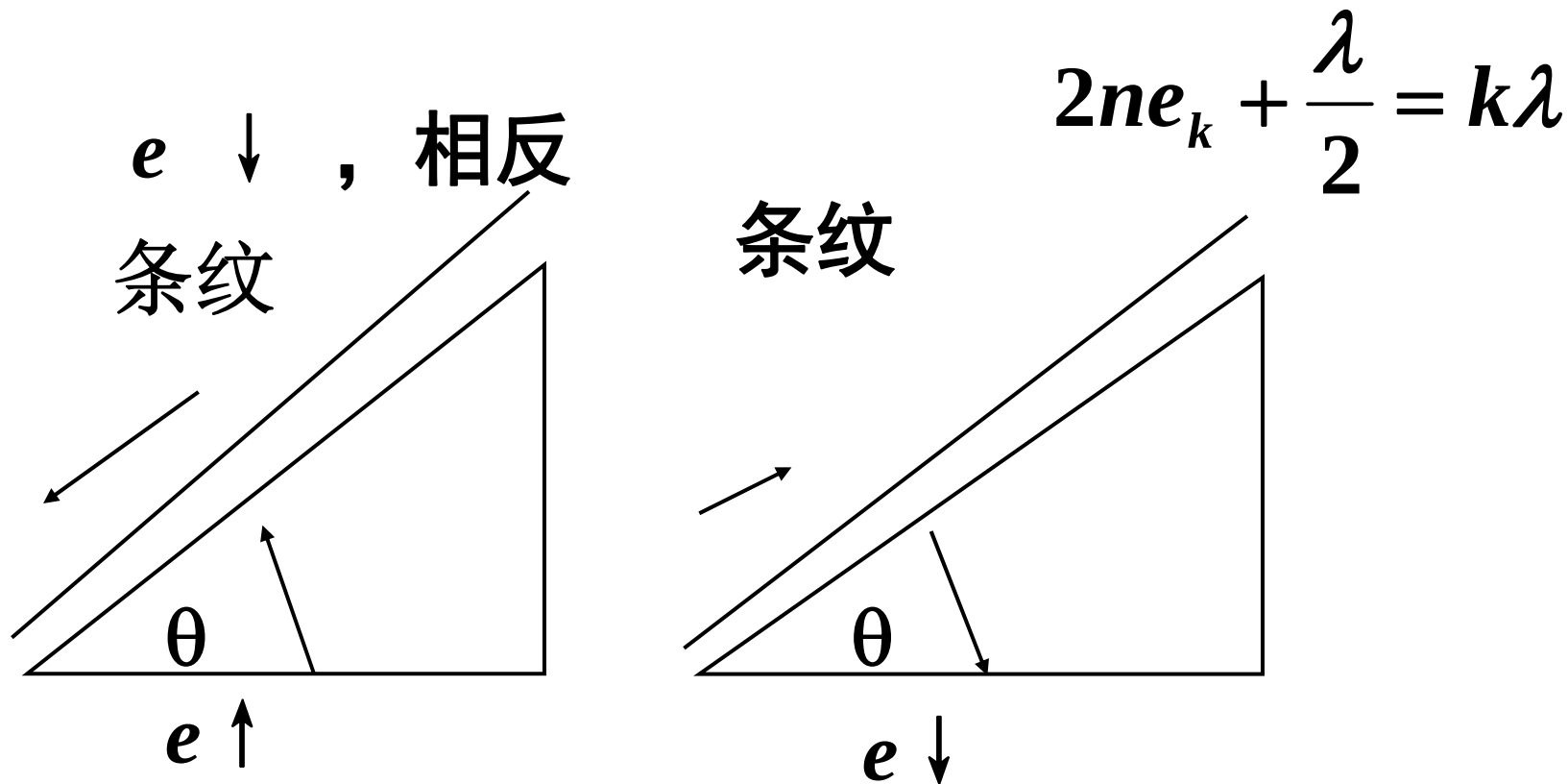
$$\delta = 2ne + \frac{\lambda}{2}$$



(1) n 、 λ 不变, $\theta \downarrow$ $L \uparrow$

条纹向远离棱边的方向运动

(2) e 均匀上升,等厚线向棱边移动



应用

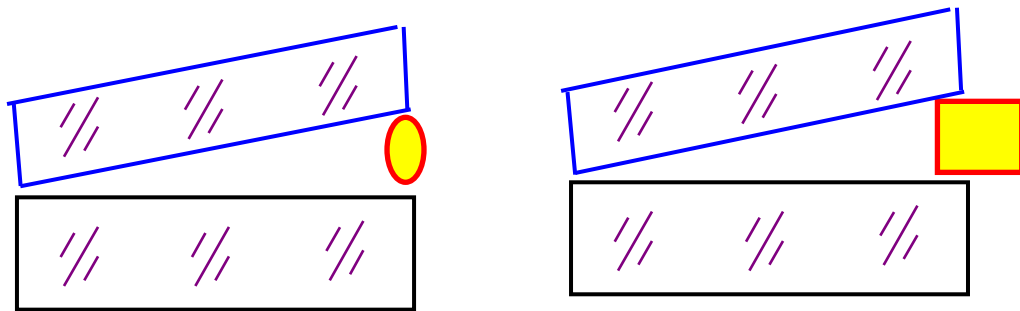
$$L = \frac{\lambda}{2n\theta}$$

1)精确测量:

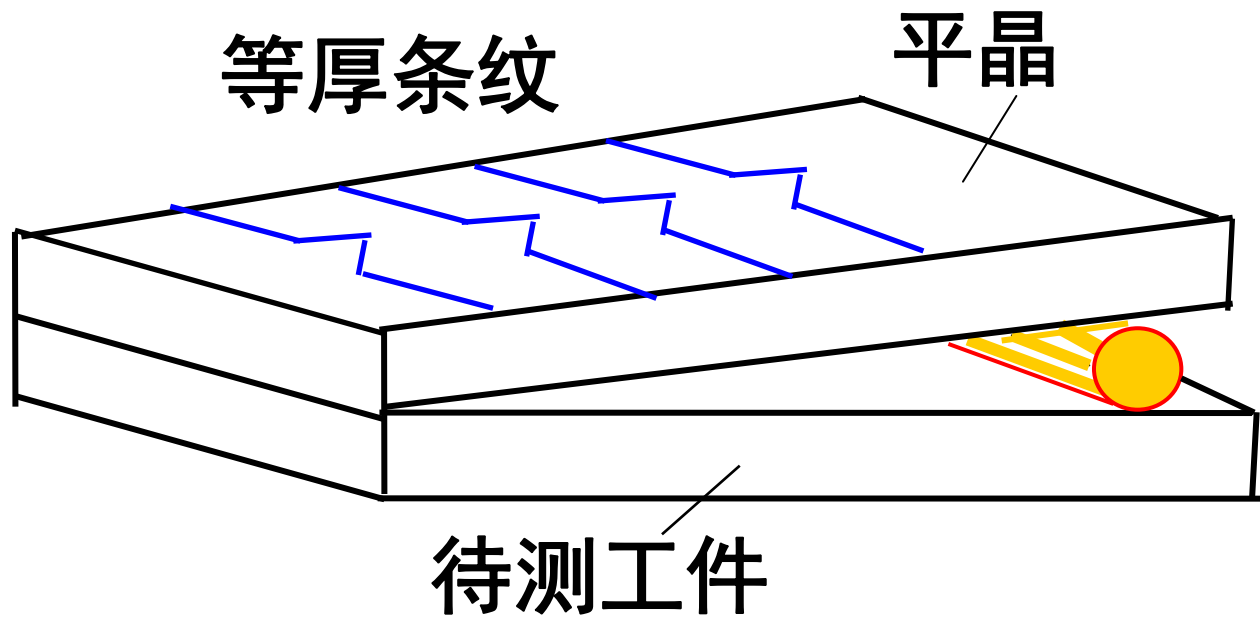
▲测波长: 已知 θ n , 测 L 可得 λ

▲测折射率: 已知 θ λ , 测 L 可得 n

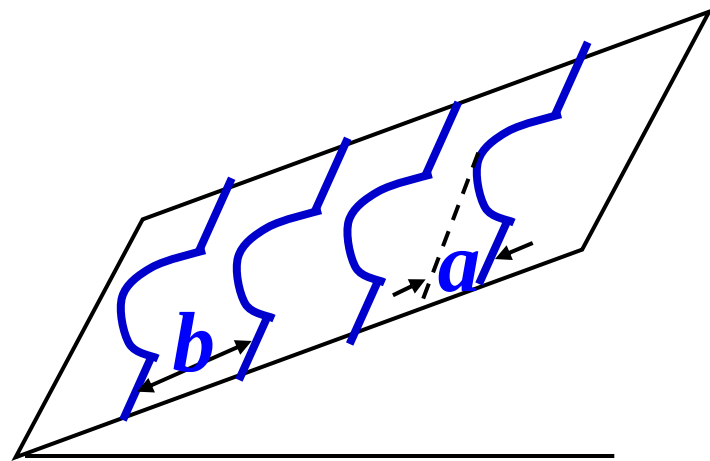
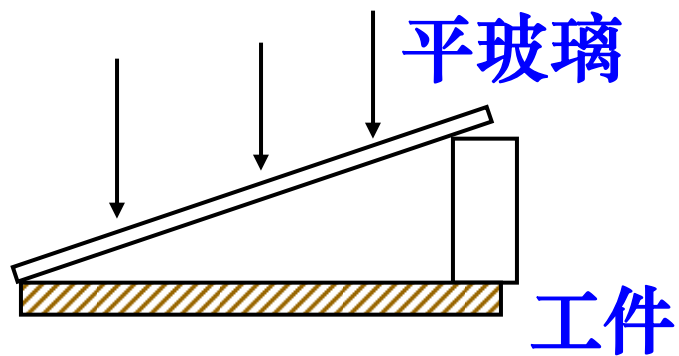
▲测细小直径、厚度、微小变化



2) 检验光学玻璃质量



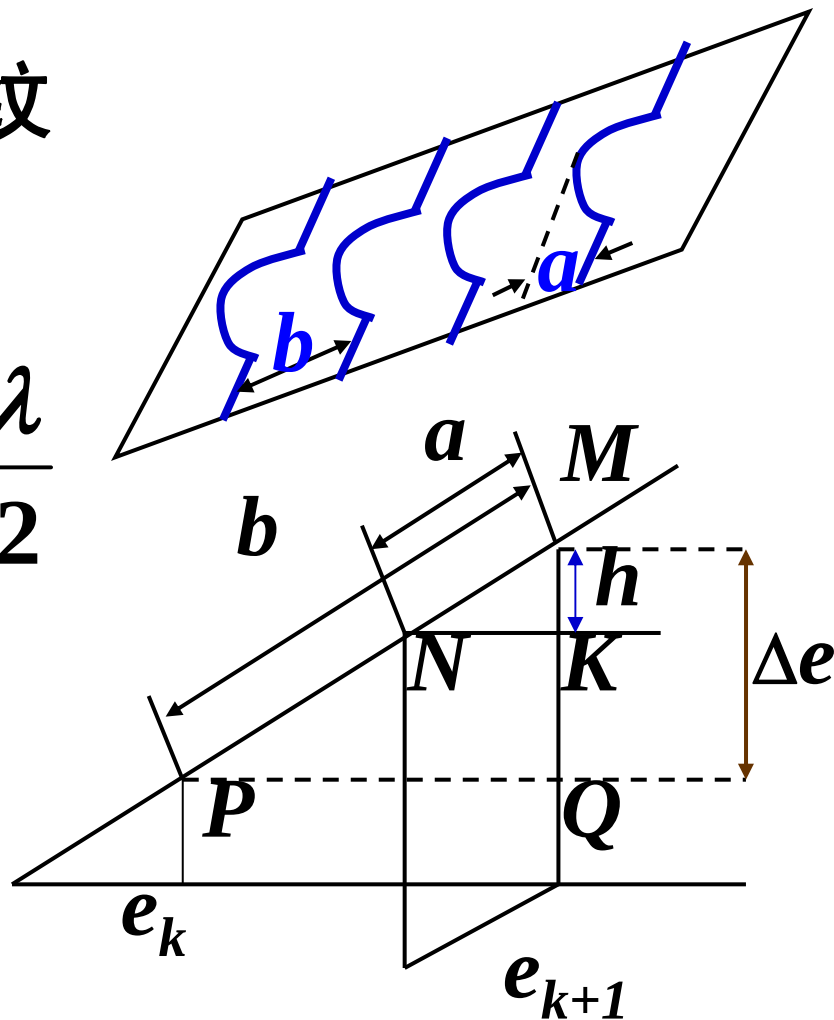
[例]: 利用等厚条纹, 检查精密加工的质量
工件上放一平玻璃, 使之形成空气劈尖, 观测到干涉条纹如图, 试判断工件表面的凹凸, 求: 纹路的深度 h 。



解：相同厚度，同一条纹
故工件下凹

$$\frac{h}{\Delta e} = \frac{a}{b} \quad \Delta e = \frac{\lambda}{2n} = \frac{\lambda}{2}$$

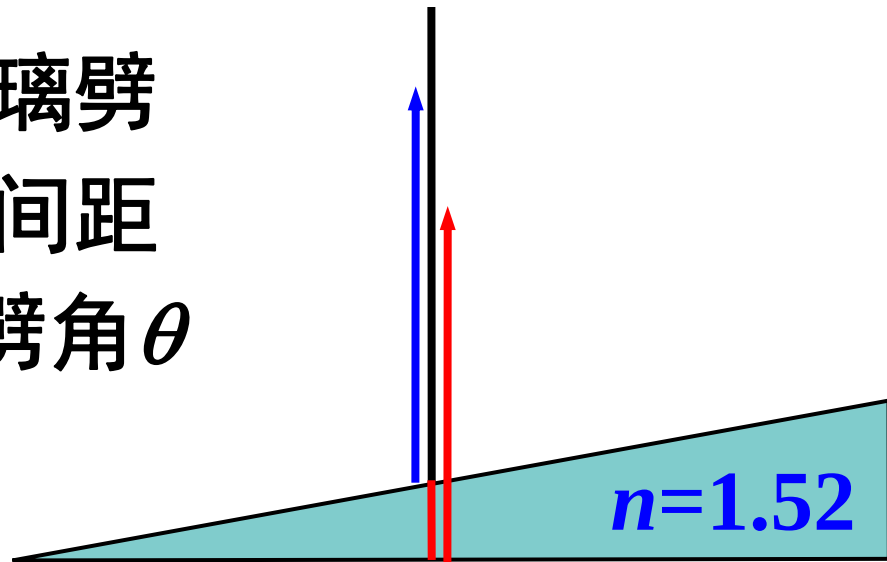
$$h = \frac{a}{b} \Delta e = \frac{a\lambda}{2b}$$



[例]: **测微小角度**: 玻璃劈, 折射率 $n=1.52$, $\lambda=5893\text{\AA}$

的钠光垂直照射玻璃劈
，测得相邻暗纹之间距
离为 $L=0.25\text{cm}$ 求劈角 θ

解:



$$\delta_k = 2ne_k + \frac{\lambda}{2} = (2k + 1) \frac{\lambda}{2} \quad (1)$$

$$\delta_k = 2ne_k + \frac{\lambda}{2} = (2k + 1)\frac{\lambda}{2} \quad (1)$$

$$\delta_{k+1} = 2ne_{k+1} + \frac{\lambda}{2} = [2(k + 1) + 1]\frac{\lambda}{2} \quad (2)$$

$$\Delta e = e_{k+1} - e_k = \frac{\lambda}{2n} \qquad \Delta e = L \sin \theta = \frac{\lambda}{2n}$$

$$\sin \theta = \frac{\lambda}{2nL} = 7.75 \times 10^{-5}$$

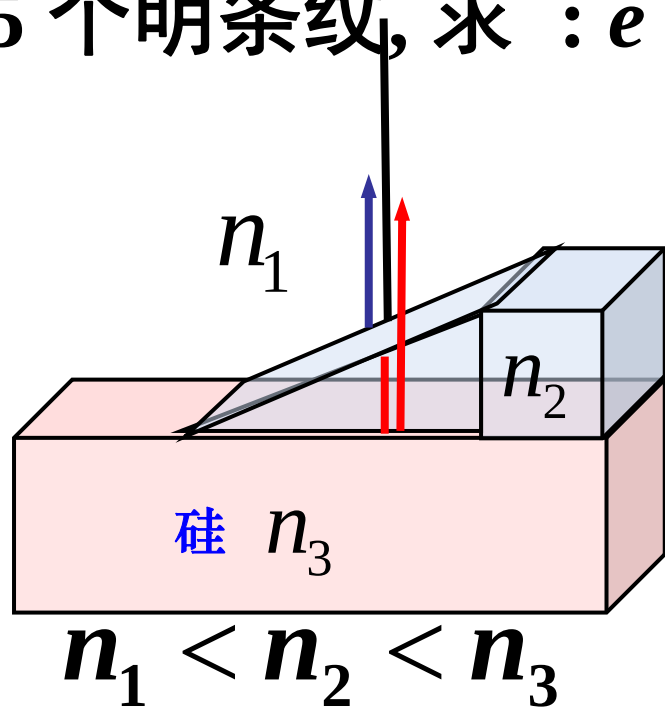
$$\theta \approx 7.75 \times 10^{-5} [\text{rad}]$$

[例]: 硅 (半导体元件中) 表面有一层氧化硅 ($n_2=1.5$), 测其厚度 e , 削成斜面, 用钠光灯 ($\lambda=5893\text{\AA}$)垂直照射看到5 个明条纹, 求 : $e = ?$

解: $\delta = 2n_2e = k\lambda$

$k=0,1,2,\dots$ 明纹

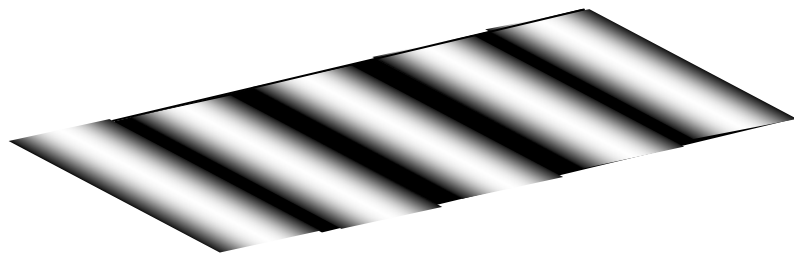
二次半波损失, 所以光程差中无 $\lambda/2$, 底边($e=0$)为明纹.



$$\delta = 2n_2e = k\lambda$$

看到5 个明条纹

取 $k=4$



$$e = \frac{k\lambda}{2n_2} = \frac{4 \times 5893 \times 10^{-10}}{2 \times 1.5} = 7.857 \times 10^{-7} [\text{m}]$$

$$e = 0.78 [\mu\text{m}]$$